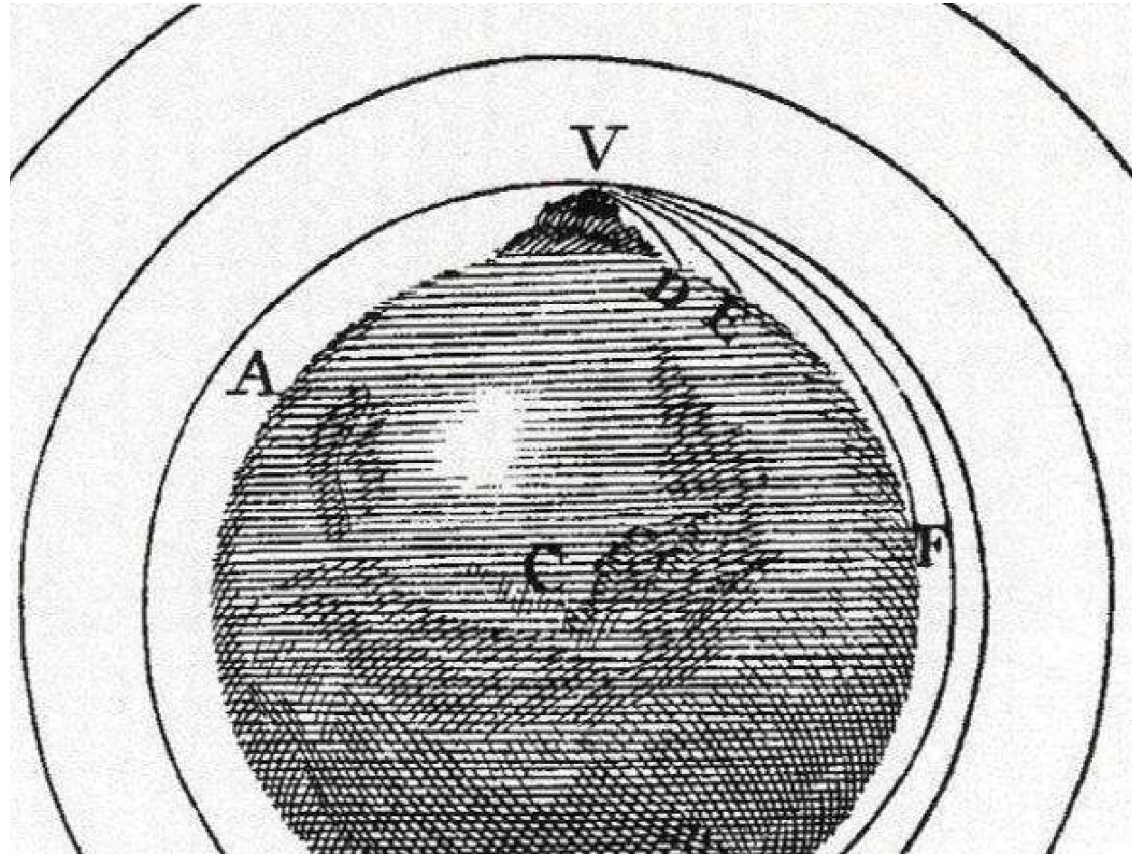


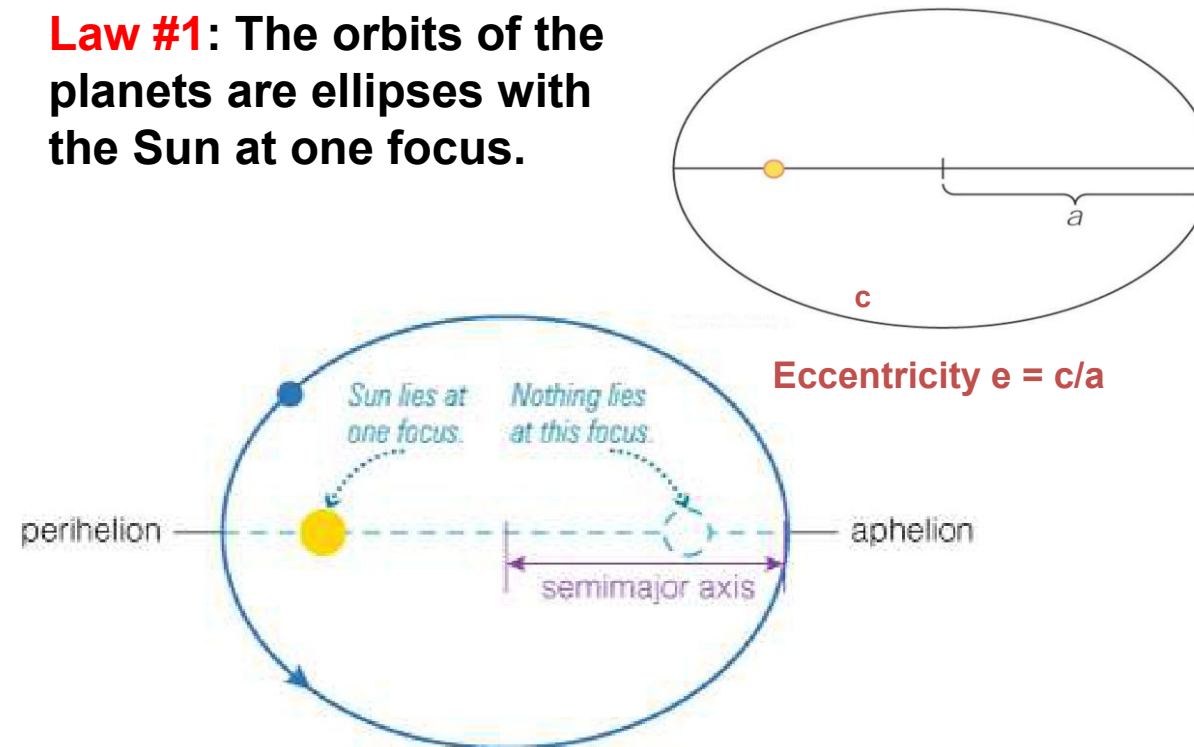
# Gravity



# Kepler's first law

## Kepler's 1<sup>st</sup> Law

**Law #1:** The orbits of the planets are ellipses with the Sun at one focus.

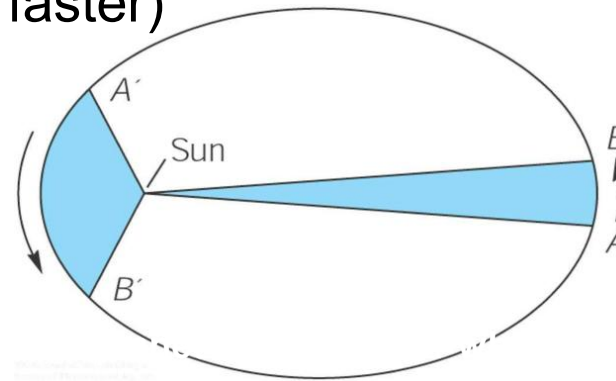


# Kepler's second law

## Kepler's 2<sup>nd</sup> Law

**Law #2** : A line from a planet to the sun sweeps over equal areas in equal intervals of time.

(When it's closer to the Sun, a planet moves faster)



# Kepler's third law

## Kepler's 3<sup>rd</sup> Law

A planet's orbital period (P) squared is proportional to its average distance from the sun (a) cubed:

**Closer orbits go faster**

$$P_y^2 = a_{AU}^3$$

( $P_y$  = period in years;  $a_{AU}$   
= distance in AU)

<http://www.solarsystemscope.com/>

[http://astro.unl.edu/naap/pos/animations/  
kepler.html](http://astro.unl.edu/naap/pos/animations/kepler.html)

## Concept Quiz

Which planet will orbit in the smallest amount of time?

- a) Something with the mass of Jupiter at the orbital distance of Jupiter
- b) Something with the mass of Earth at the orbital distance of Jupiter
- c) Something with the mass of Earth at the orbital distance of Earth
- d) Something with the mass of Jupiter at the orbital distance of Earth
- e) c) and d) would be equal

## Concept Quiz

Which planet will orbit in the smallest amount of time?

- a) Something with the mass of Jupiter at the orbital distance of Jupiter
- b) Something with the mass of Earth at the orbital distance of Jupiter
- c) Something with the mass of Earth at the orbital distance of Earth
- d) Something with the mass of Jupiter at the orbital distance of Earth
- e) c) and d) would be equal

## Concept Quiz

How long does it take Mars to orbit the Sun?

- a) More than a year
- b) Less than a year
- c) One year

## Concept Quiz

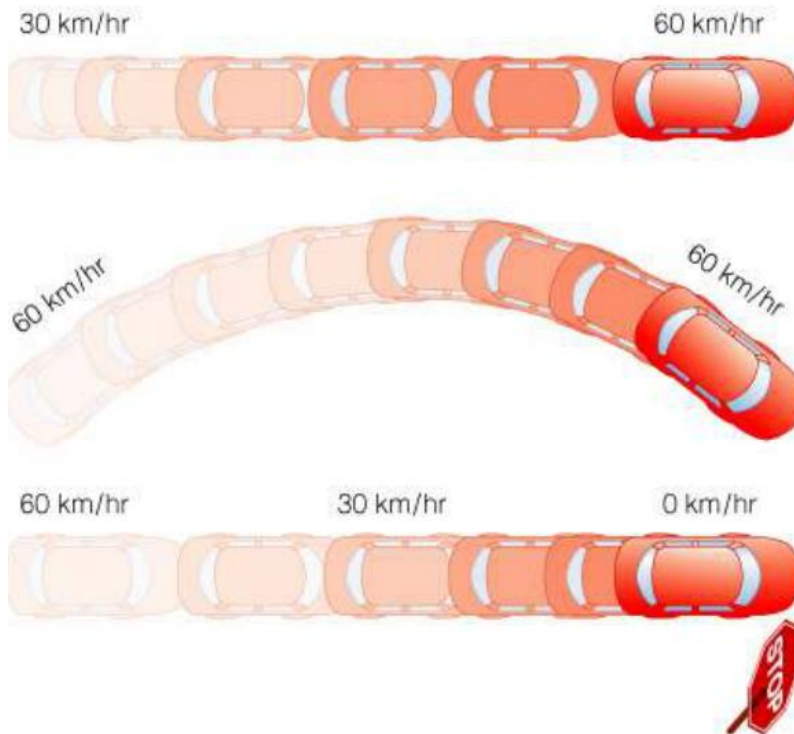
How long does it take Mars to orbit the Sun?

- a) More than a year
- b) Less than a year
- c) One year

# Goals for Today

- 1) What is gravity?
- 2) What is proportionality?
- 3) What is the inverse square law?

# How do we describe motion?



**Speed** : The rate at which an object moves  
example: 10 m/s

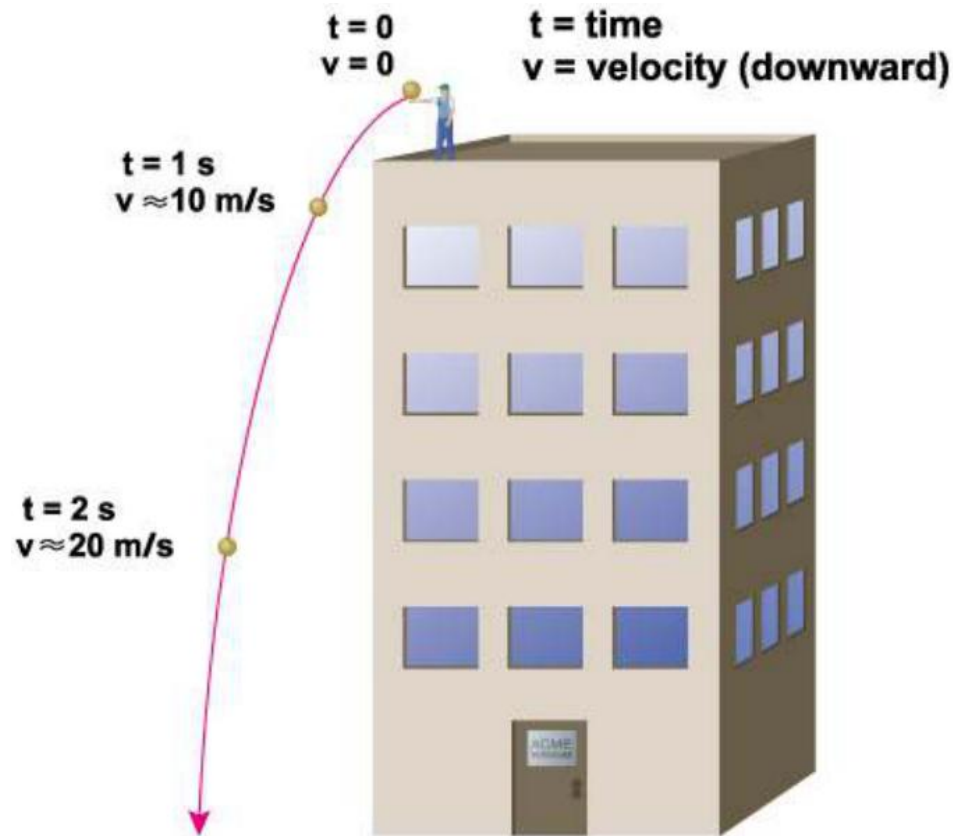
**Velocity** : Speed + direction  
example: 10 m/s, due east

**Acceleration** : Any change in velocity, including a change in direction.

# Gravity is an Acceleration!

On Earth, speed increases 10 m/s with each second of falling time.

This increase in speed is the acceleration due to gravity



# The Acceleration due to Gravity

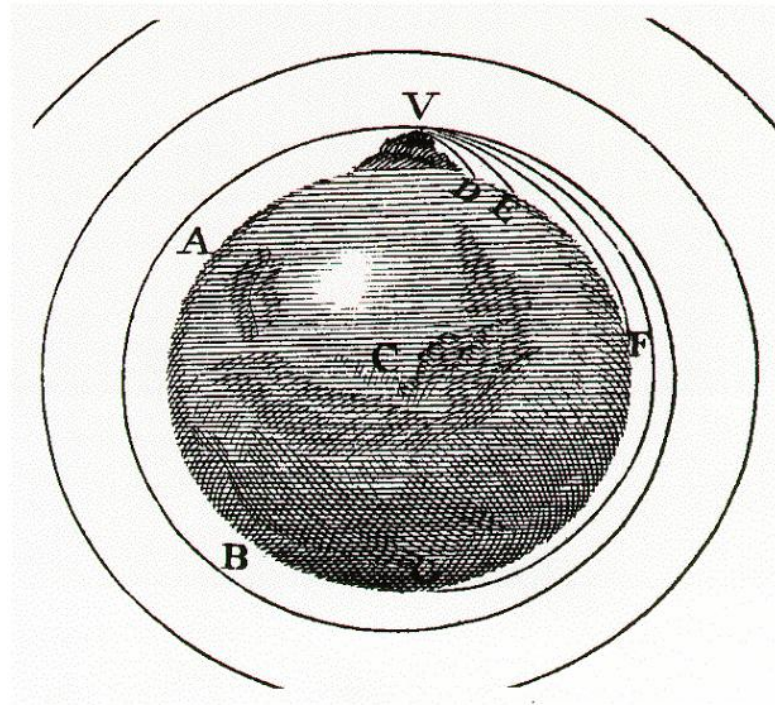
Galileo showed that the gravitational acceleration is the same for all falling objects, regardless of their mass.

Air resistance, not an object's mass, is what causes some objects to fall slower than others.



[http://www.youtube.com/watch?v=-4\\_rceVPVSY](http://www.youtube.com/watch?v=-4_rceVPVSY)

# Why are astronauts “weightless” in space?



There *is* gravity  
in space

Weightlessness  
is due to a  
constant state of  
free-fall

# The Universal Law of Gravitation

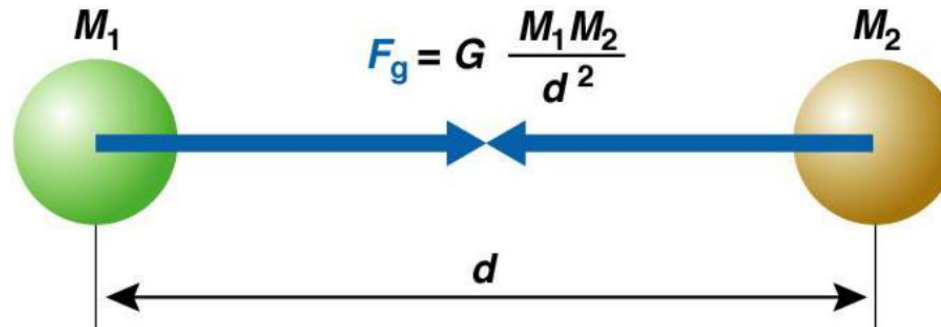
Every mass attracts every other mass.

Attraction is directly proportional to the product of their masses.

– As the masses increase, so does the gravitational force

Attraction is inversely proportional to the square of the distance between their centers.

– As the distances decrease, the gravitational force increases quickly



# Proportionality

When two terms are proportional, as one increases the other one increases

Terms can be inversely proportional: as one increases the other decreases.

**Important!!!**

## Concept Quiz

The force of gravity is proportional to \_\_\_\_\_ and inversely proportional to \_\_\_\_\_.

- a) acceleration, distance between objects
- b) distance between objects, product of object masses
- c) product of object masses, acceleration
- d) product of object masses, distance between objects

# Concept Quiz

The force of gravity is proportional to \_\_\_\_\_ and inversely proportional to \_\_\_\_\_.

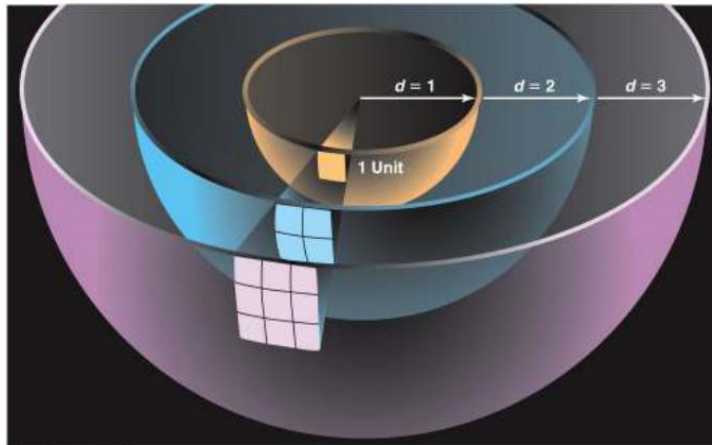
- a) acceleration, distance between objects
- b) distance between objects, product of object masses
- c) product of object masses, acceleration
- d) product of object masses, distance between objects

# Inverse Square Law

An **inverse square law** is a relation where the quantity decreases with the square of the distance

OR: Intensity decreases with distance, quickly

Examples: Light, gravity, and sound



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## Concept Quiz

If the Earth were moved to 2AU, how would the force of gravity between it and the Sun change?

- a) Impossible to know
- b) Decrease by a factor of 2
- c) Increase by a factor of 4
- d) Decrease by a factor of 4

## Concept Quiz

If the Earth were moved to 2AU, how would the force of gravity between it and the Sun change?

- a) Impossible to know
- b) Decrease by a factor of 2
- c) Increase by a factor of 4
- d) Decrease by a factor of 4

Twice as far away = 4 times less ( $2^2$ )

## Concept Quiz

If the Earth were moved to 0.5AU, how would the force of gravity between it and the Sun change?

- a) Impossible to know
- b) Decrease by a factor of 2
- c) Increase by a factor of 4
- d) Decrease by a factor of 4

## Concept Quiz

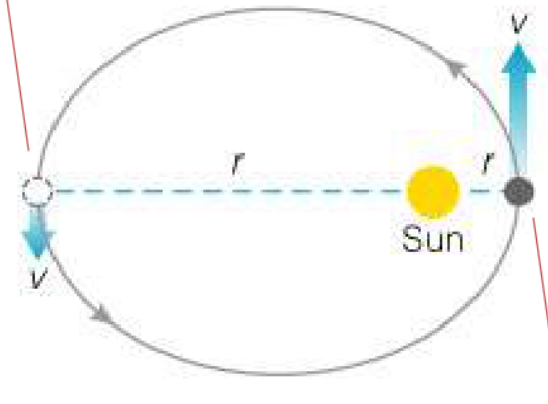
If the Earth were moved to 0.5AU, how would the force of gravity between it and the Sun change?

- a) Impossible to know
- b) Decrease by a factor of 2
- c) Increase by a factor of 4
- d) Decrease by a factor of 4

Half as far away = 4 times greater  
 $(0.5^{-2}) = 4$

# Gravity and Orbits

More gravitational energy;  
Less kinetic energy



Less gravitational energy;  
More kinetic energy

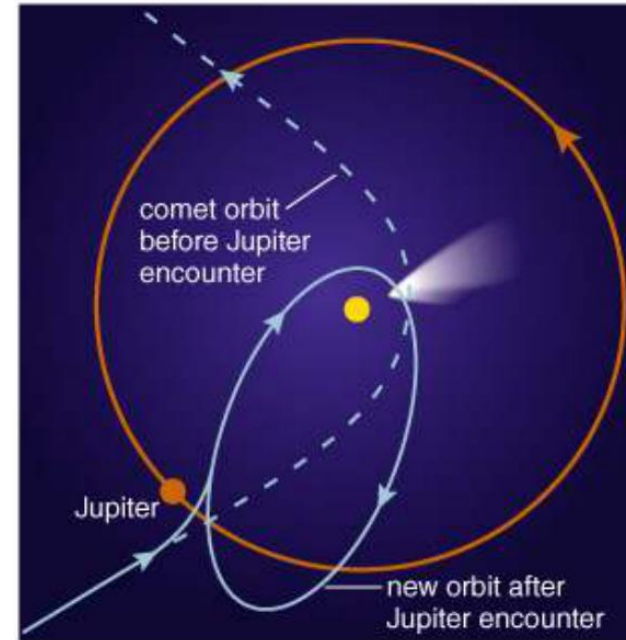
Total orbital  
energy  
(gravitational +  
kinetic) stays  
constant if  
there is no  
external force  
Orbits cannot  
change  
spontaneously.

# Gravity

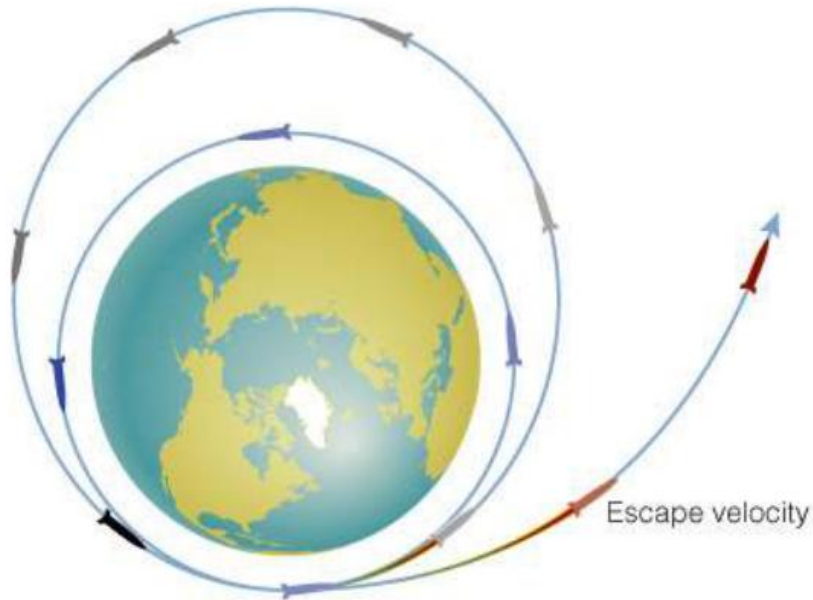


# Changing an Orbit

- ⇒ So what can make an object gain or lose orbital energy?
- Friction or atmospheric drag
    - Reduces velocity
    - Decreases kinetic energy
    - Orbit decreases in size
  - A gravitational encounter.
    - Energy can transfer to the other body



# Escape Velocity



If an object gains enough orbital energy, it may escape (change from a bound to unbound orbit)

**Escape velocity** from Earth  $\approx 11$  km/s from sea level (about 40,000 km/hr)

# What have we learned?

## 1) What is gravity?

Gravity is an acceleration due to the gravitational force.

## 2) What is proportionality?

Two terms are proportional if an increase in one quantity leads to an increase in the other. They are inversely proportional if an increase in one quantity leads to a decrease in the other.

## 3) What is an inverse square law?

An inverse square law is a relation where the quantity decreases with the square of the distance